Course 6: Present and Future of AI

IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom

Everything in this session (reasoning models, agents, music generators) is built on the ingredients of session 5: a Transformer trained by self supervision, then adapted.

Present and future of AI

The present

- Assistants, reasoning models, **agents** acting on a computer,
- Models that generate **music**, images, video,
- Measured costs: **electricity**, money, dependence.

The future

- Not a prediction: **choices** being made now,
- By companies, states, artists… and engineers,
- Three open questions, each with **two sides**.

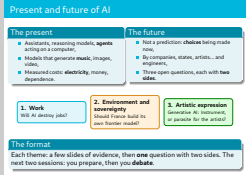
1. Work
Will AI destroy jobs?

2. Environment and sovereignty
Should France build its own frontier model?

3. Artistic expression
Generative AI: Instrument, or parasite for the artists?

The format

Each theme: a few slides of evidence, then **one** question with two sides. The next two sessions: you prepare, then you **debate**.



The questions are written so that a serious person can defend either side. If one of them looks obviously settled to you, you might be surprised by the viewpoints of your peers.

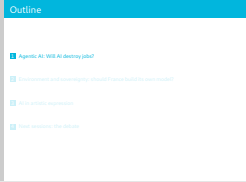
Outline

1 Agentic AI: Will AI destroy jobs?

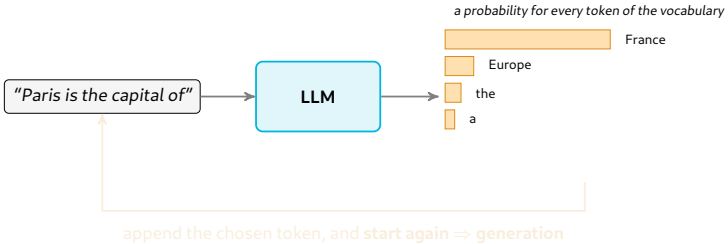
2 Environment and sovereignty: should France build its own model?

3 AI in artistic expression

4 Next sessions: the debate



Using a Large Language Model (LLM) to generate text

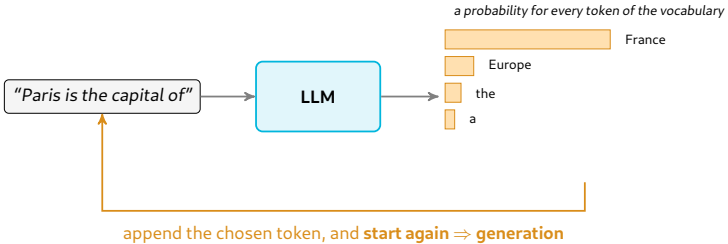




The next token is *sampled* from these probabilities, not always the most probable one: this is why the same prompt can give different answers.

Each new token requires a full pass through the model: the cost of generation grows with the length of the text produced.

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From text completion to instruction following

The problem

Pre-training teaches the model to **continue text**, not to **answer**.

Before (raw pre-trained)

You: What is the capital of France?

Model: What is the capital of Germany? What is the capital of Spain? …

it just keeps the pattern going

After (instruction-tuned)

You: What is the capital of France?

Model: Paris.

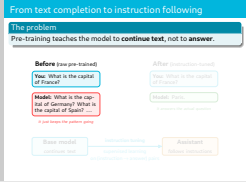
it answers the actual question

Base model — *continues* text

instruction tuning

supervised learning on (instruction → answer) pairs

Assistant — *follows* instructions



The "before" box is actually very relevant: on the internet, a question is very often followed by other questions, so a list of questions is the correct answer to the only problem the model was trained on.

Instruction tuning uses tens of thousands of demonstrations, written or curated by humans (summarizing, translating, explaining, refusing, formatting). The model learns no new knowledge here (it already knew the capital of France), only the **format** of an answer.

The limit: somebody has to *write* the ideal answer. Feasible for "what is the capital of France", much less so for "explain quantum computing to a child" or "is this request dangerous".

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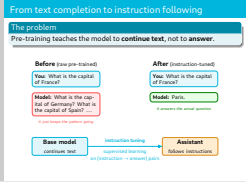
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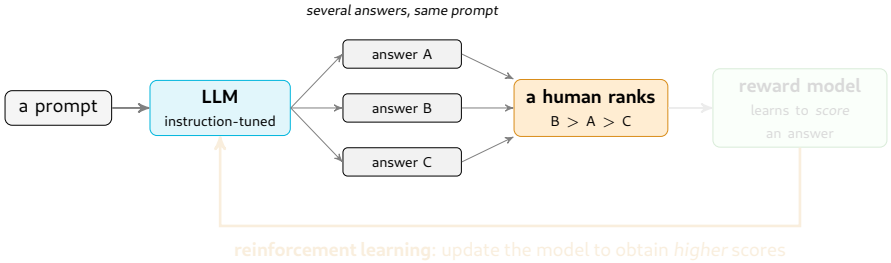
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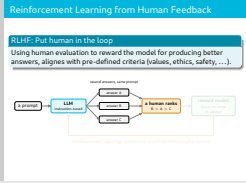
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Reinforcement Learning from Human Feedback

RLHF: Put human in the loop

Using human evaluation to reward the model for producing better answers, alignes with pre-defined criteria (values, ethics, safety, …).





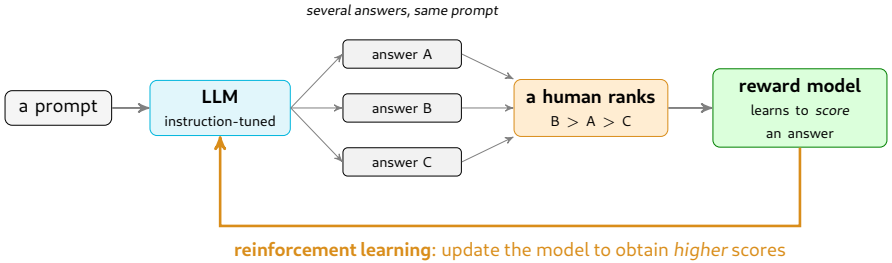
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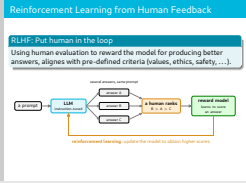
The several answers come from sampling (previous slides). The reward model is what makes the method scalable: once trained, it judges millions of answers without any human. The gradient no longer comes from a ground truth, it comes from a judge. (Reinforcement learning itself is not covered in this course.)

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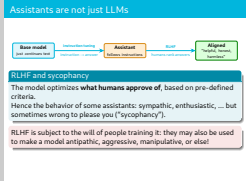
Assistants are not just LLMs



RLHF and sycophancy

The model optimizes **what humans approve of**, based on pre-defined criteria.
Hence the behavior of some assistants: sympathic, enthusiastic, ... but sometimes wrong to please you ("sycophancy").

RLHF is subject to the will of people training it: they may also be used to make a model antipathic, aggressive, manipulative, or else!



Human approval correlates with quality, but is not the same thing: a confident, well formatted, agreeable answer is rated higher, including when it is wrong, and an answer that contradicts the user is generally rated lower. Keep it in mind when you use an assistant: it was trained to be *approved of*.

Reasoning models: Chain-of-Thought

Giving time to think

Taking time to write a reasoning process before answering.
Leads to better answers on arithmetic, logic, and planning problems, but increases the computation cost.

Direct answer

Q: I have 5 balls, and I buy 6 cans of 7 balls. How many balls?

A: 43 ×

it guesses in one shot

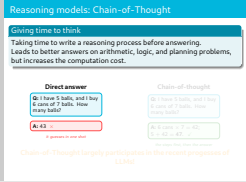
Chain-of-thought

Q: I have 5 balls, and I buy 6 cans of 7 balls. How many balls?

A: 6 cans × 7 = 42;
5 + 42 = **47**. ✓

the steps first, then the answer

Chain-of-Thought largely participates in the recent progesses of LLMs!



Why it works: the model has a fixed amount of computation per token. Written intermediate steps become part of the context, so the problem is cut into pieces small enough for one pass each. Nothing is hidden and no module is added: the model is simply given a scratchpad (think of a street interview versus a written exam).

"Reasoning models" are trained (with the preference methods of the previous slides) to produce this chain of thought by themselves, and to make it longer when the problem is hard.

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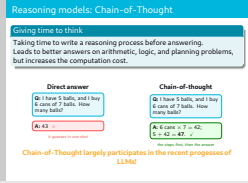
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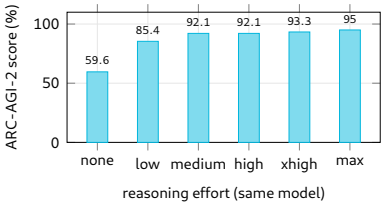
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CoT performance, and its cost

Two places to spend compute

Training: paid once, the model improves for every user.
Inference (CoT): paid at **every question**, and you choose how much.



Longer reasoning improves accuracy!

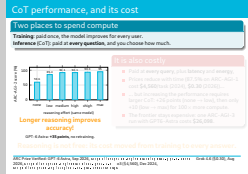
GPT-6 Astra: **+35 points**, no retraining.

It is also costly

- Paid at **every query**, plus **latency** and **energy**,
- Prices reduce with time (87.5% on ARC-AGI-1 cost **$4,560**/task (2024), **$0.30** (2026))...
- ... but increasing the performance requires larger CoT: +26 points (none → low), then only +10 (low → max) for 100× more compute.
- The frontier stays expensive: one ARC-AGI-3 run with GPT6-Astra costs **$26,098**.

Reasoning is not free: its cost moved from training to every answer.

ARC Prize Verified: GPT-6 Astra, Sep 2026, `https://arcprize.org/results/openai-gpt-6-astra` Grok 4.6 ($0.30), Aug 2026, `https://arcprize.org/results/xai-grok-4-6` o3 ($4,560), Dec 2024, `https://arcprize.org/blog/oai-o3-pub-breakthrough`



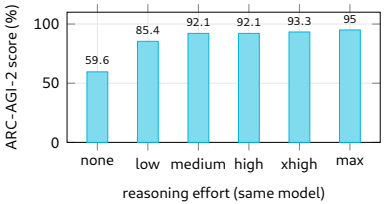
ARC-AGI-2: visual puzzles designed to be easy for humans and hard for machines. ARC-AGI-3 is its newer, interactive version.
CoT is very efficient, but also expensive!

Why this is important in the context of the "jobs question": an answer now has a **price per task**. A task done at a few thousand dollars cannot be industrially automated, whatever the benchmark says; while a task done at a few cents may be.

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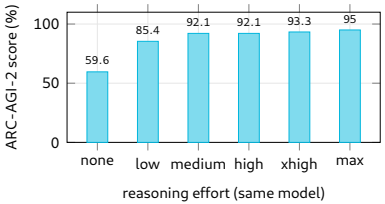
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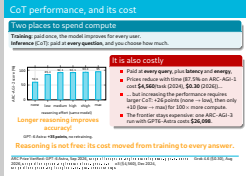
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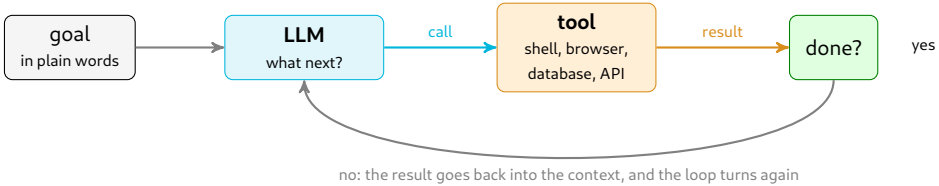
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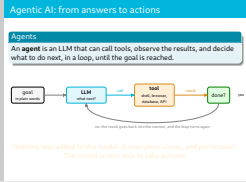
Agentic AI: from answers to actions

Agents

An **agent** is an LLM that can call tools, observe the results, and decide what to do next, in a loop, until the goal is reached.



Nothing was added to the model. It was given a loop, and permission.
The model is now able to take actions!



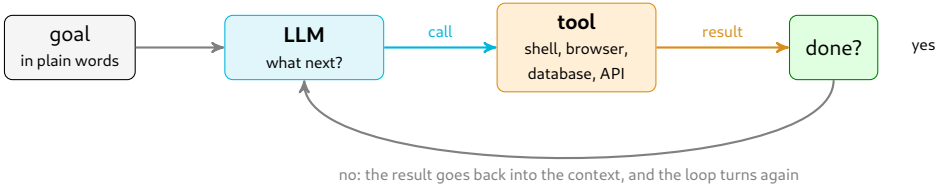
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In practice, the model never executes anything itself: it writes a structured request (e.g. which file to read), and the program around it executes it and pastes the result back into the context. So the risks of an agent come less from the model than from the **permissions** it is given (reading your mail, running commands, paying).

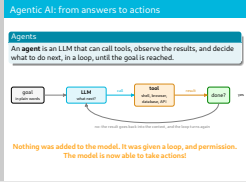
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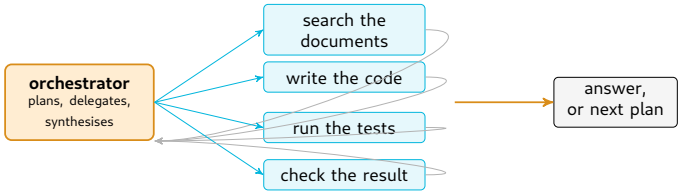
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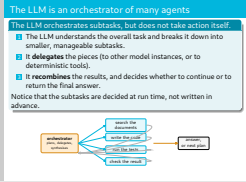
The LLM is an orchestrator of many agents

The LLM orchestrates subtasks, but does not take action itself.

1. The LLM understands the overall task and breaks it down into smaller, manageable subtasks.
2. It **delegates** the pieces (to other model instances, or to deterministic tools).
3. It **recombines** the results, and decides whether to continue or to return the final answer.

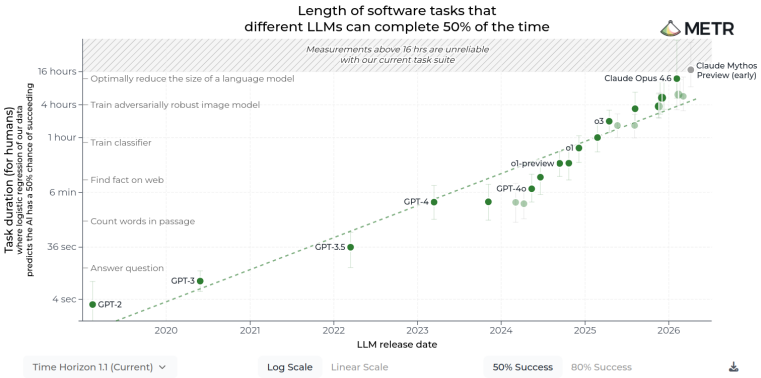
Notice that the subtasks are decided at run time, not written in advance.





The cost: every worker is a full model call, and the orchestrator reads all their outputs. An agentic query can cost an order of magnitude more energy and money than a chat query.

Agents can perform longer and longer tasks

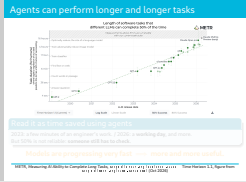


Read it as time saved using agents

2023: a few minutes of an engineer's work. / 2026: a **working day**, and more.
But 50% is not reliable: **someone still has to check**.

Models are progressing very fast $\Longrightarrow$ more and more useful.

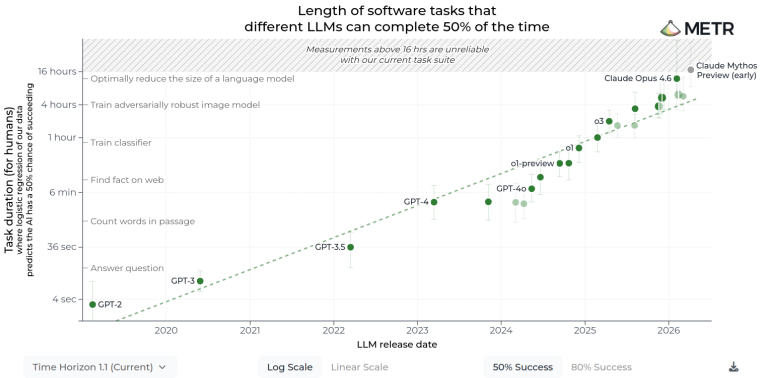
METR, *Measuring AI Ability to Complete Long Tasks*, `https://arxiv.org/abs/2503.14499` Time Horizon 1.1, figure from `https://metr.org/time-horizons/` (Oct 2026)



Method: METR times skilled humans on a few hundred real tasks (from a few seconds to more than a day of work), then finds the task length an agent completes half of the time. In early 2026, Claude Opus 4.6 reached about twelve hours (uncertainty: between five and sixty hours). Since 2023, the horizon doubled about every four months (129 days); since 2019, about every six months.

Three cautions. At 80% success, horizons are five to ten times shorter (about one hour for Claude Opus 4.6), and errors **compound** on long tasks (99% success per step gives about 5% success over 300 steps). The tasks are software, machine learning and cybersecurity: do not transfer the curve to other domains. The last grey point (Claude Mythos Preview) is beyond what the task suite can measure: METR excludes it from the trend.

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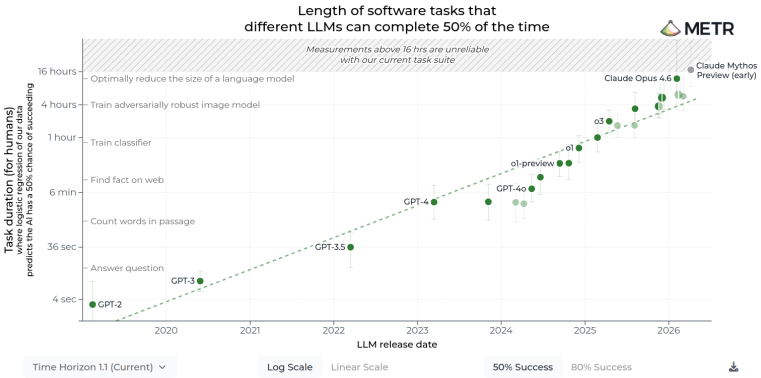
Models are progressing very fast $\Longrightarrow$ more and more useful.

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Method: METR times skilled humans on a few hundred real tasks (from a few seconds to more than a day of work), then finds the task length an agent completes half of the time. In early 2026, Claude Opus 4.6 reached about twelve hours (uncertainty: between five and sixty hours). Since 2023, the horizon doubled about every four months (129 days); since 2019, about every six months. Three cautions. At 80% success, horizons are five to ten times shorter (about one hour for Claude Opus 4.6), and errors **compound** on long tasks (99% success per step gives about 5% success over 300 steps). The tasks are software, machine learning and cybersecurity: do not transfer the curve to other domains. The last grey point (Claude Mythos Preview) is beyond what the task suite can measure: METR excludes it from the trend.

Agents can perform longer and longer tasks

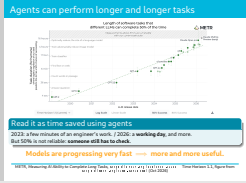


Read it as time saved using agents

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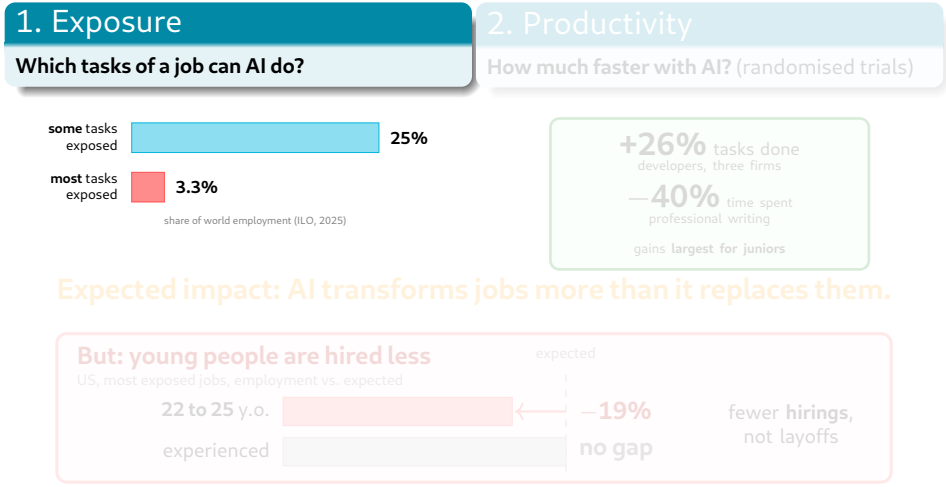
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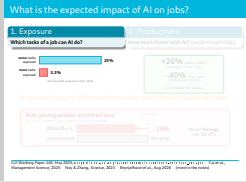


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What is the expected impact of AI on jobs?



ILO Working Paper 140, May 2025, https://www.ilo.org/sites/default/files/2025-05/WP140_web.pdf Cui et al., *Management Science*, 2025 Noy & Zhang, *Science*, 2023 Brynjolfsson et al., Aug 2026 (more in the notes)



A historical counterweight first: about 60% of US jobs in 2018 were in job titles that did not exist in 1940 (Autor et al., QJE 2024); new work appears, but over decades.

Exposure is computed by splitting each occupation into tasks, and scoring whether a language model could do each one. Headlines often merge the two figures (25% and 3.3%). It is not gender neutral: clerical work is the most exposed category, and mostly done by women (9.6% of women's jobs against 3.5% of men's in high income countries).

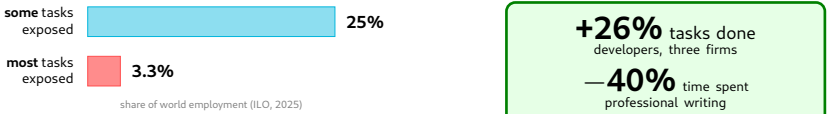
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Which tasks of a job can AI do?



2. Productivity

How much faster with AI? (randomised trials)

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Expected impact: AI transforms jobs more than it replaces them.

But: young people are hired less

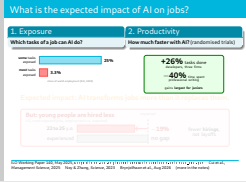
US, most exposed jobs, employment vs. expected

22 to 25 y.o. ← −19%

experienced — no gap

fewer **hirings**, not layoffs

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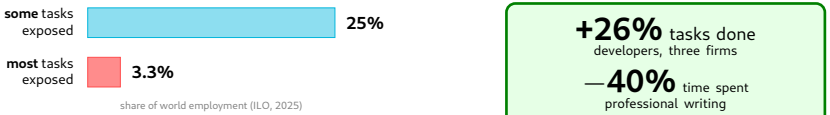
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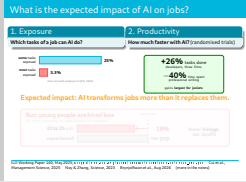
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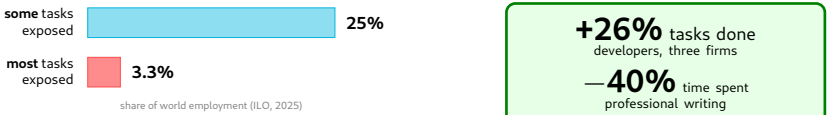
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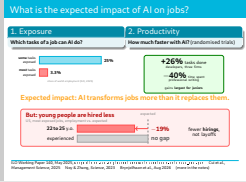
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Place yourself: will AI destroy jobs?

pick one side

YES *or* **NO**

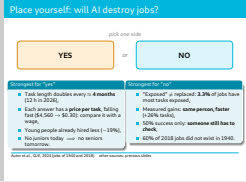
Strongest for "yes"

- Task length doubles every ≈ **4 months** (12 h in 2026),
- Each answer has a **price per task**, falling fast ($4,560 → $0.30): compare it with a wage,
- Young people already hired less (−19%),
- No juniors today ⟹ no seniors tomorrow.

Strongest for "no"

- "Exposed" ≠ replaced: **3.3%** of jobs have most tasks exposed,
- Measured gains: **same person, faster** (+26% tasks),
- 50% success only: **someone still has to check**,
- 60% of 2018 jobs did not exist in 1940.

Autor et al., *QJE*, 2024 (jobs of 1940 and 2018) other sources: previous slides



The question hides two. **Can it do the task?** An empirical question, partly answered in this section. **Will it be deployed to replace the person?** A question about prices, management and law, that no benchmark answers: a productivity gain can become shorter hours, cheaper goods, higher wages, fewer hires or higher profits.

On juniors: the lower rungs of a profession (the junior lawyer reading contracts, the junior developer fixing small bugs) are economically weak but pedagogically essential. If they become cheaper to automate than to teach, the ladder loses its bottom, and nobody has an answer to that yet.

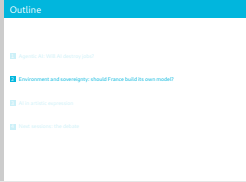
Outline

1 Agentic AI: Will AI destroy jobs?

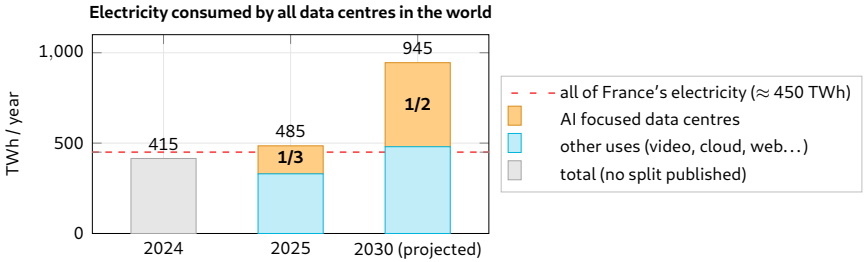
2 Environment and sovereignty: should France build its own model?

3 AI in artistic expression

4 Next sessions: the debate



What AI costs today, and where it is going



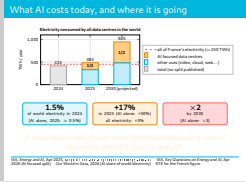
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(AI alone, 2025: ≈ 0.5%)

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all electricity: +3%

×2
by 2030
(AI alone: ×3)

A small share today; but it's one of the fastest growing uses of electricity, and AI drives the growth.

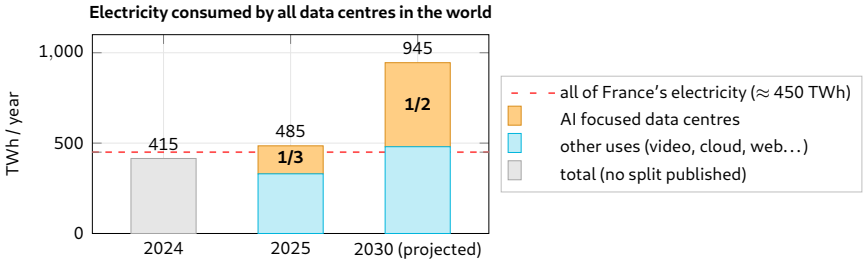
IEA, *Energy and AI*, Apr 2025, https://www.iea.org/reports/energy-and-ai (AI focused split) Our World in Data, 2026 (AI share of world electricity) IEA, *Key Questions on Energy and AI*, Apr 2026 RTE for the French figure



The International Energy Agency (IEA) publishes the AI split from 2025 only. Its 2030 projections are uncertain, and the IEA is usually considered rather conservative.

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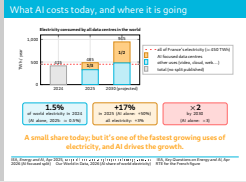
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One model, measured: Mistral Large 2

Building the model
training, then 18 months of service

20,400 t CO_2e
≈ *the yearly footprint of 2,200 French people*

a large cost, for the model as a whole

+

One more answer
Le Chat, 400 tokens
1.14 g CO_2e

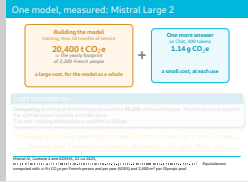
a small cost, at each use

Cost breakdown

Computing (training and inference) account for **85.5%** of the emissions. Mistral does not publish the split between training and inference.
The rest: making the hardware and the buildings.

Training and using one Mistral model costs as much as a small town… And we don't know for OpenAI, Anthropic, Google, Meta, etc.

Mistral AI, Carbone 4 and ADEME, 22 Jul 2025,
`https://mistral.ai/fr/news/our-contribution-to-a-global-environmental-standard-for-ai/` Equivalences computed with ≈ 9 t CO_2e per French person and per year (SDES) and 2,500 m^3 per Olympic pool



The first full lifecycle analysis published by a model developer, made with an independent consultancy (Carbone 4) and the French environment agency (ADEME), following a public French methodology (AFNOR, "frugal AI"). Its authors call it a first approximation, mainly because the manufacturing footprint of GPUs is poorly documented.
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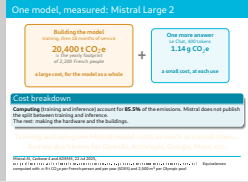
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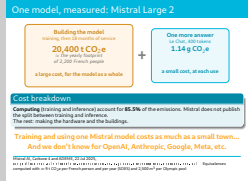
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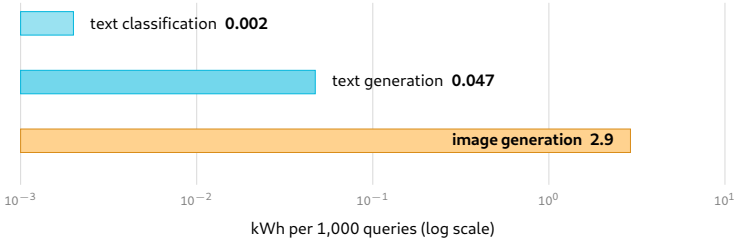
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Not all tasks cost the same



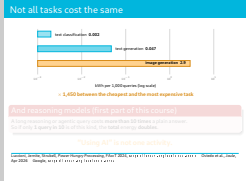
× **1,450 between the cheapest and the most expensive task**

And reasoning models (first part of this course)

A long reasoning or agentic query costs **more than 10 times** a plain answer.
So if only **1 query in 10** is of this kind, the **total** energy **doubles**.

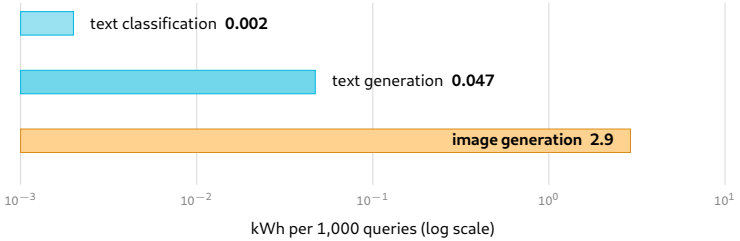
"Using AI" is not one activity.

Luccioni, Jernite, Strubell, *Power Hungry Processing*, FAccT 2024, https://arxiv.org/abs/2311.16863 Oviedo et al., *Joule*, Apr 2026 Google, https://arxiv.org/abs/2508.15734



For reference, the best documented figure for a plain text prompt is Google's: a median of **0.24 Wh** (2025).

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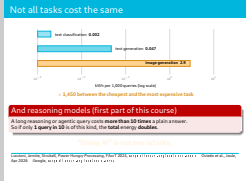
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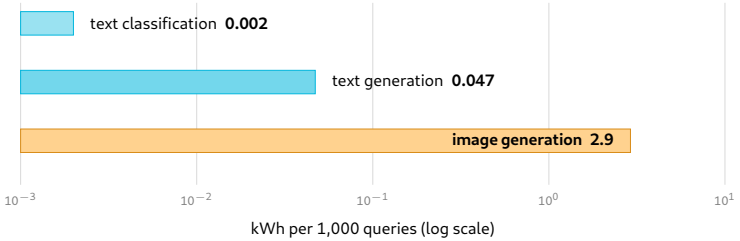
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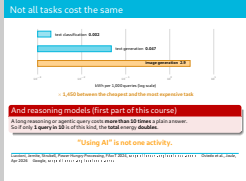
× 1,450 between the cheapest and the most expensive task

And reasoning models (first part of this course)

A long reasoning or agentic query costs **more than 10 times** a plain answer.
So if only **1 query in 10** is of this kind, the **total** energy **doubles**.

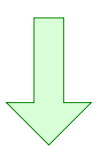
"Using AI" is not one activity.

Luccioni, Jernite, Strubell, *Power Hungry Processing*, FAccT 2024, https://arxiv.org/abs/2311.16863 Oviedo et al., *Joule*, Apr 2026 Google, https://arxiv.org/abs/2508.15734



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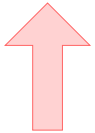
Computing efficiency, and the rebound effect



÷ 33

energy **per prompt** at Google, in 12 months

but



+37%

Google **total** electricity, in 2025 (emissions: +81% since 2019)

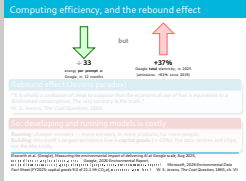
Rebound effect (Jevons paradox)

"It is wholly a confusion of ideas to suppose that the economical use of fuel is equivalent to a diminished consumption. The very contrary is the truth."
W. S. Jevons, *The Coal Question*, 1865

So: developing and running models is costly

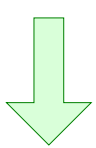
Running: cheaper answers → more answers, in more products, for more people.
Building: Microsoft's largest emissions line is **capital goods** (≈ 43%): the data centres and chips, not the electricity.

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Jevons observed that more efficient steam engines did not reduce Britain's coal consumption: coal power became cheaper, so it was used everywhere. Economists call this the **rebound effect** (or Jevons paradox). Efficiency arguments therefore cannot stand alone in a debate about a growing technology.

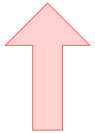
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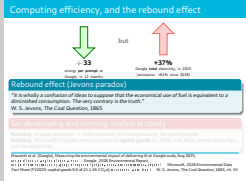
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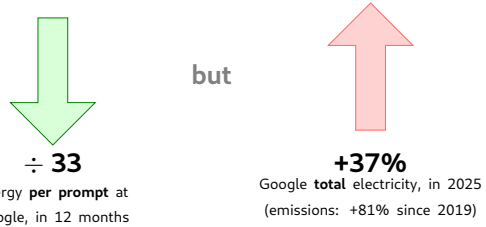
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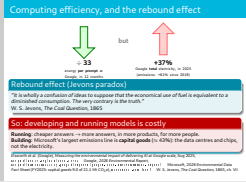
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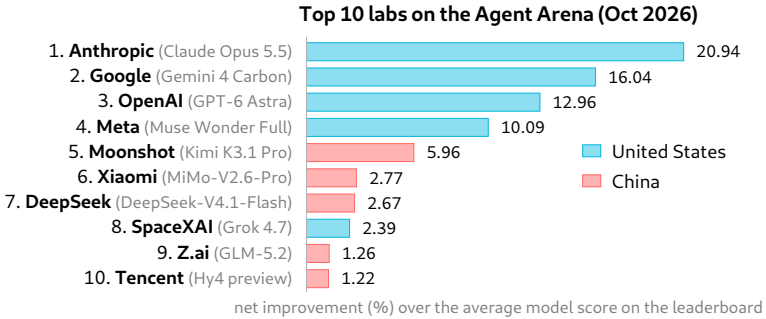
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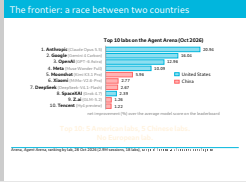
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The frontier: a race between two countries



Top 10: 5 American labs, 5 Chinese labs.
No European lab.

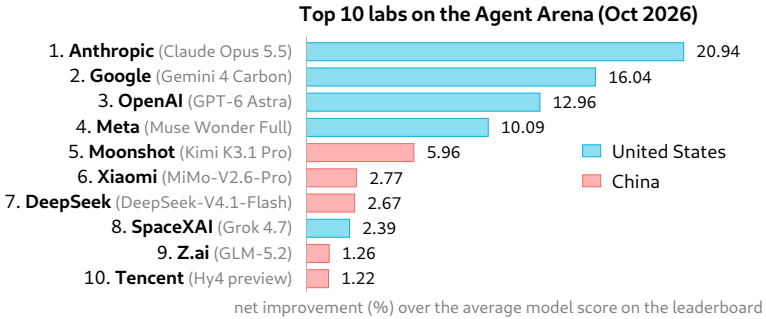
Arena, *Agent Arena*, ranking by lab, 28 Oct 2026 (2.9M sessions, 18 labs), `https://arena.ai/leaderboard/agent`



The “frontier” is the set of the most capable models at a given time. The Agent Arena ranks them from almost 3 million real sessions where people use them as agents, rather than from a fixed benchmark. “Net improvement” combines task success, user satisfaction and steerability.

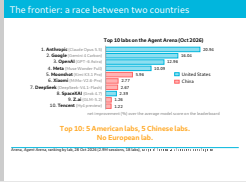
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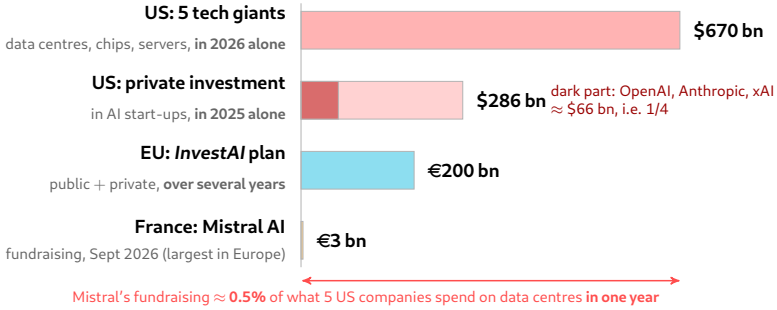
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A ferocious and very costly competition

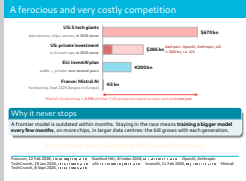


Why it never stops

A frontier model is outdated within months. Staying in the race means **training a bigger model every few months**, on more chips, in larger data centres: the bill grows with each generation.

France (and even Europe) is left behind (for now).

Futurum, 12 Feb 2026, `futurumgroup.com` Stanford HAI, *AI Index 2026*, `hai.stanford.edu` OpenAI, Anthropic: TechCrunch, 19 Jan 2026, `techcrunch.com` xAI: `techfundingnews.com` InvestAI, 11 Feb 2025, `maginative.com` Mistral: TechCrunch, 8 Sept 2026, `techcrunch.com`



The bars are drawn to scale. Details:

US: 5 tech giants (Amazon, Alphabet/Google, Microsoft, Meta, Oracle): announced *capital expenditure* (buildings, chips, servers, power connections), about $660 to 690 billion according to their own guidance in early 2026.

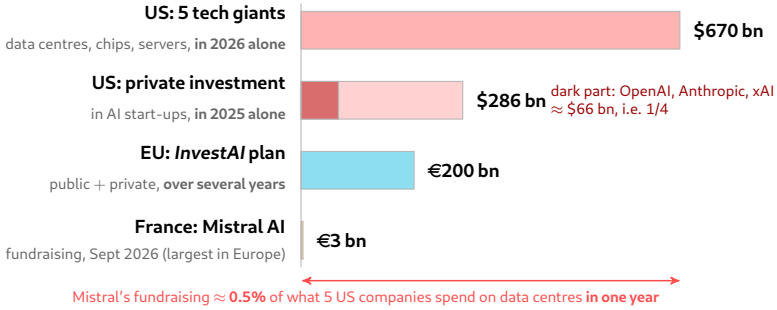
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Mistral AI: most recent fundraising round, Sept 2026, largest in Europe.

Being second at the frontier is worth much less than being first, hence the race.

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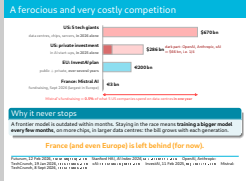


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Sovereignty is not only the AI model!

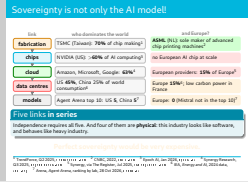
link	who dominates the world	and Europe?
fabrication	TSMC (Taiwan): **70%** of chip making[1]	**ASML** (NL): sole maker of advanced chip printing machines[2]
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cloud	Amazon, Microsoft, Google: **63%**[4]	European providers: **15%** of Europe[5]
data centres	US **45%**, China 25% of world consumption[6]	Europe **15%**[6]; low carbon power in France
models	Agent Arena top 10: US **5**, China **5**[7]	Europe: **0** (Mistral not in the top 10)[7]

Five links **in series**

Independence requires all five. And four of them are **physical**: this industry looks like software, and behaves like heavy industry.

Perfect sovereignty would be very expensive.

[1] TrendForce, Q2 2025, `trendforce.com` [2] CNBC, 2022, `cnbc.com` [3] Epoch AI, Jan 2026, `epoch.ai` [4] Synergy Research, Q3 2025, `srgresearch.com` [5] Synergy, via The Register, Jul 2025, `theregister.com` [6] IEA, *Energy and AI*, 2024 data, `iea.org` [7] Arena, *Agent Arena*, ranking by lab, 28 Oct 2026, `arena.ai`



Most advanced chips are made by one company (TSMC), on one island (Taiwan). NVIDIA's dominance is also its software (CUDA), which almost all AI code is written against.
ASML makes the machines every advanced chip factory needs, so this is a real leverage. Nonetheless, they don't manufacture chips!

Low carbon electricity in France is an argument to host data centres, but connecting a new site to the grid takes years.

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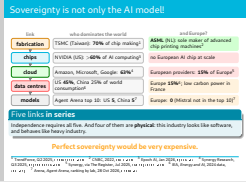
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Already dependent: Europe runs on American Big Tech



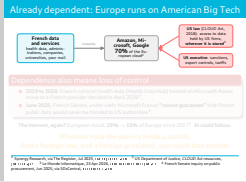
Dependence also means loss of control

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The Internet, again? European cloud: **29%** → **15%** of Europe since 2017[1]. **AI could follow.**

Whoever runs the service holds a switch.
And a foreign law, and a foreign president, can reach that switch.

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Since 2017, the European cloud market grew sixfold, and almost all of the growth went to three American firms.

The CLOUD Act lets US authorities require an American company to hand over data it controls, even on servers in Europe: "hosted in France" is not "protected by French law". The political lever is not hypothetical either: in 2025, after US sanctions against its prosecutor, the International Criminal Court decided to leave Microsoft for a European open source suite (Computing, 31 Oct 2025).

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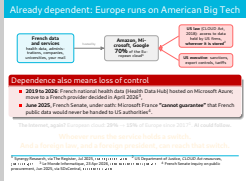
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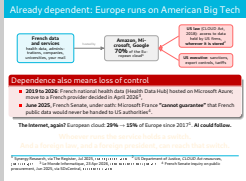
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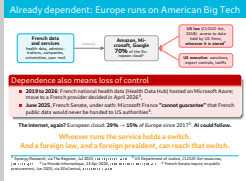
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Place yourself: should France build its own frontier model?

pick one side

YES *or* **NO**

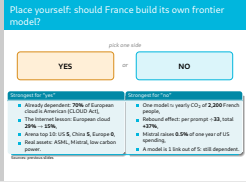
Strongest for "yes"

- Already dependent: **70%** of European cloud is American (CLOUD Act),
- The Internet lesson: European cloud **29%** → **15%**,
- Arena top 10: US **5**, China **5**, Europe **0**,
- Real assets: ASML, Mistral, low carbon power.

Strongest for "no"

- One model ≈ yearly CO_2 of **2,200** French people,
- Rebound effect: per prompt ÷**33**, total **+37%**,
- Mistral raises **0.5%** of one year of US spending,
- A model is 1 link out of 5: still dependent.

Sources: previous slides



The question is a single trade: the electricity, the capital and the political attention come from the same pot.
Two refinements worth noting:
On the *yes* side: build, but not alone (pool it at European level, as with CERN for capital intensive science).

On the *no* side: do not develop a frontier model, but specialised ones on French and European data (health, industry, energy), that would reduce the environmental impact.

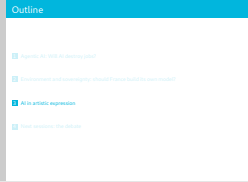
Outline

1 Agentic AI: Will AI destroy jobs?

2 Environment and sovereignty: should France build its own model?

3 AI in artistic expression

4 Next sessions: the debate

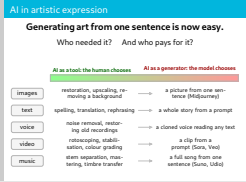


AI in artistic expression

Generating art from one sentence is now easy.

Who needed it? And who pays for it?

	AI as a tool: the human chooses	AI as a generator: the model chooses
images	restoration, upscaling, removing a background	a picture from one sentence (Midjourney)
text	spelling, translation, rephrasing	a whole story from a prompt
voice	noise removal, restoring old recordings	a cloned voice reading any text
video	rotoscoping, stabilisation, colour grading	a clip from a prompt (Sora, Veo)
music	stem separation, mastering, timbre transfer	a full song from one sentence (Suno, Udio)



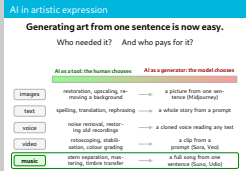
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What AI music enables

the human makes the choices — **the model makes the choices**

Restore
The Beatles, *Now and Then* (2023)
Lennon's voice **separated** from an old cassette (separation, not generation). Grammy 2025

Tools for musicians
Stems, mastering, timbres
separate the instruments of a track, master a demo, turn your voice into a violin (RAVE, IRCAM, 2021)

Give a voice back
Randy Travis (2024)
voiceless after a stroke, releases a new song: his voice recreated by AI, **with his consent**

New creators
Xania Monet (2025)
a human writes the lyrics, Suno makes the music: $3M record deal. Suno: **2 million** paying users (2026)

Some music was never meant to be art

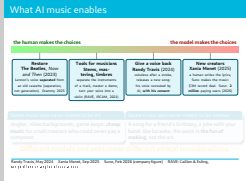
Jingles, video backgrounds, game loops: **cheap music** for small creators who could never pay a composer.

Some music was never meant to be serious

A song for a friend's birthday, a joke with your band: like karaoke, the point is **the fun of making**, not the art.

Different models and uses mean different ethical considerations.

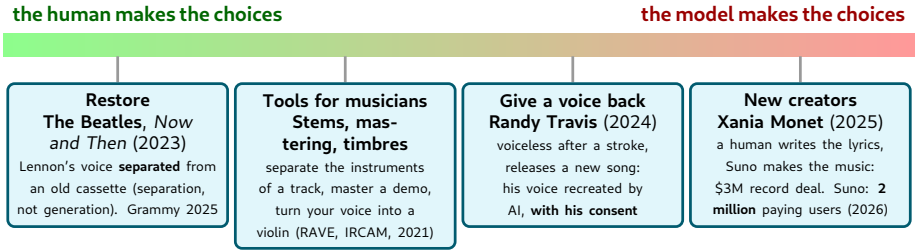
Randy Travis, May 2024 Xania Monet, Sep 2025 Suno, Feb 2026 (company figure) RAVE: Caillon & Esling, https://arxiv.org/abs/2111.05011



Now and Then is **separation**, not generation: the performance is Lennon's.
Some models, such as RAVE (IRCAM, Paris), may be used to transform (timbre here, in real time) rather than generate.
Randy Travis lost his voice after a stroke in 2013: another singer recorded the song, and a model trained on Randy Travis's own recordings turned it into his voice.
Xania Monet is the AI artist of Telisha Jones, who writes the lyrics.

The bill: every one of these uses relies on models trained on somebody's music, and on a market where somebody used to be paid for jingles. The next slides look at it.

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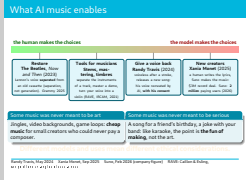
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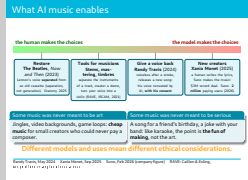
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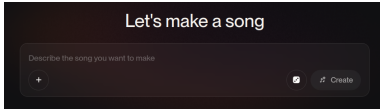
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Data and model: the example of Suno



text prompt

copyrighted music → ? → a song

No consent, no licence: training admitted in court [2024], sources re-sealed by a hack [2026]

No transparency: no paper, no model card, no data list

"But humans learn from music too!"

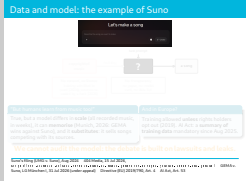
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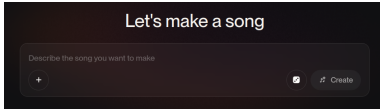


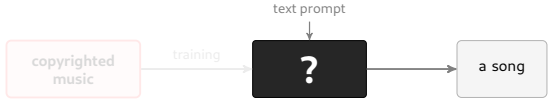
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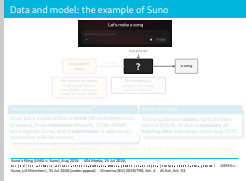
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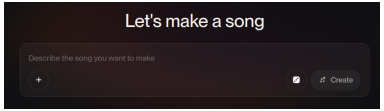


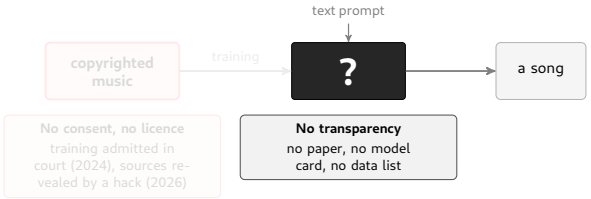
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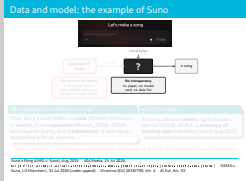
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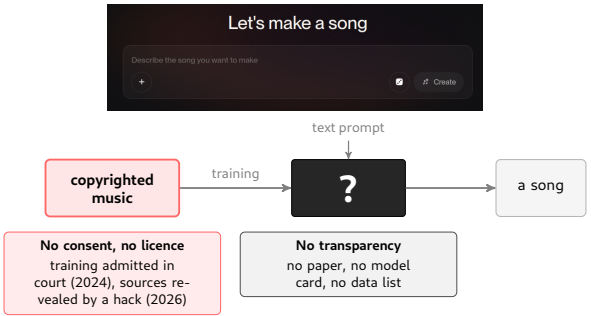


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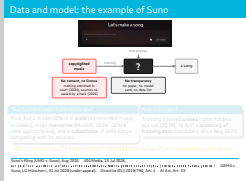
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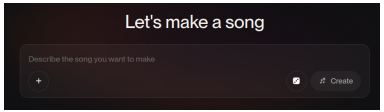


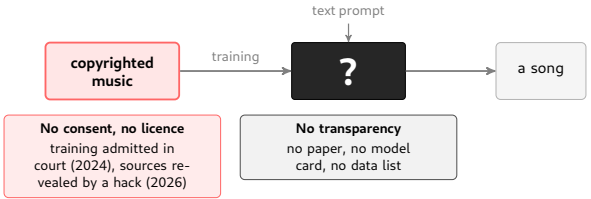
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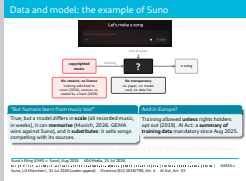
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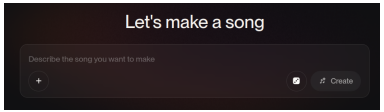
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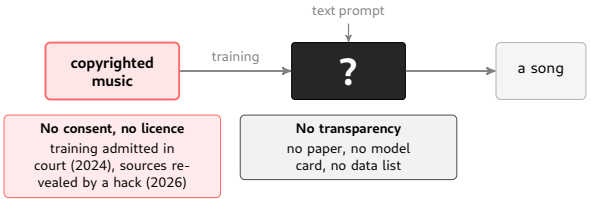
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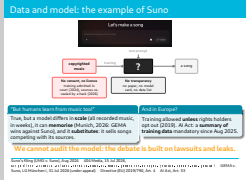
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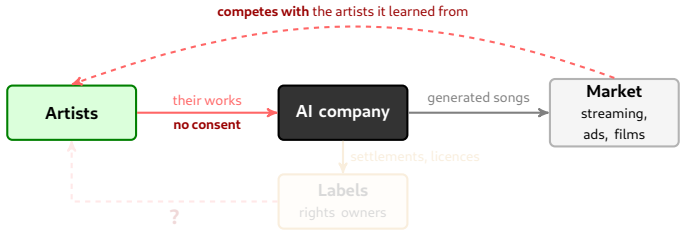
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Artists vs. the industry: who decides, who is paid?



Said yes, on their terms

Holly Herndon (2021): her voice as an open model.
Grimes (2023): use my voice, **50%** of royalties.
Warner + Suno (2025): artists **opt in**.

Said no

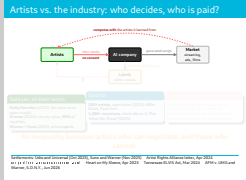
200+ artists, open letter (2024): Billie Eilish, Pearl Jam.
1,000+ musicians, silent album *Is This What We Want?* (2025).

Never asked

Fake Drake (2023): a cloned voice, millions of plays, then removed.
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An inequality between artists who can negotiate, and those who cannot.

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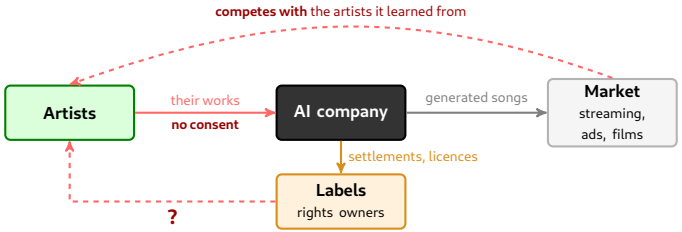


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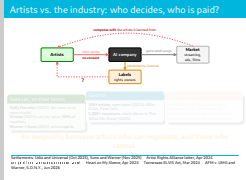
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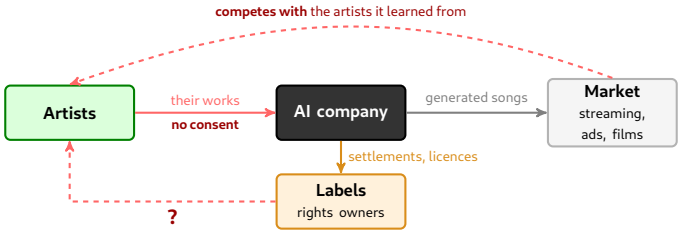


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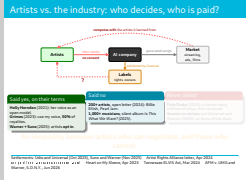
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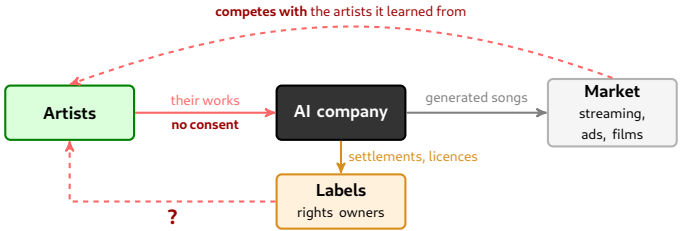


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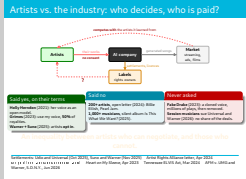
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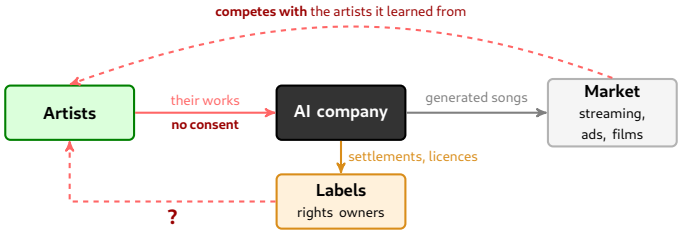


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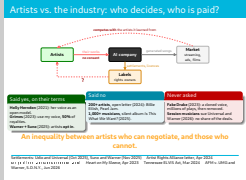
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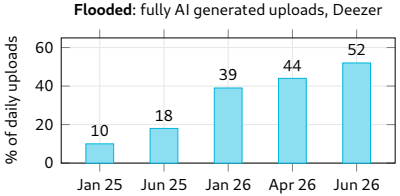


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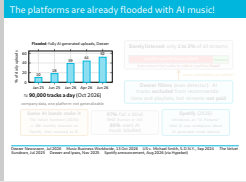
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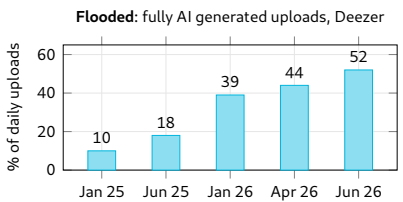
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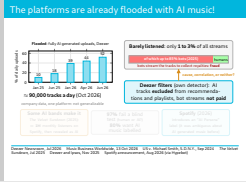
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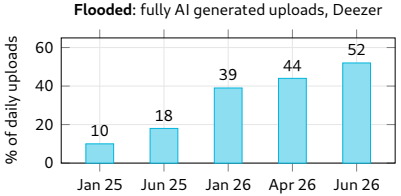
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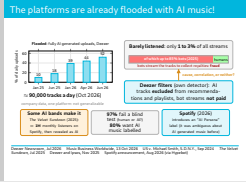
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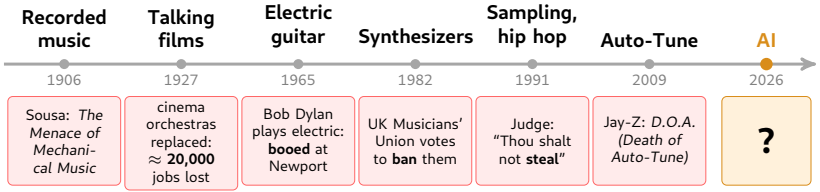
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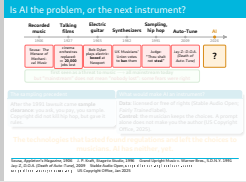
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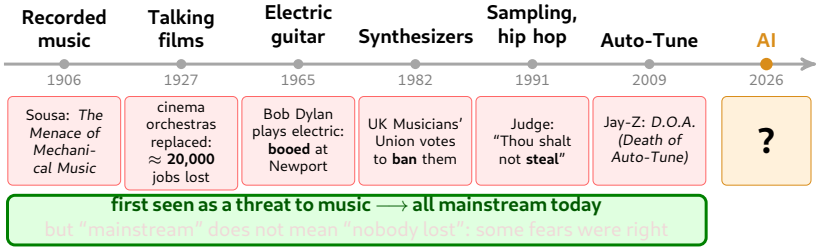
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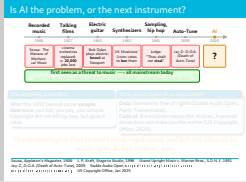
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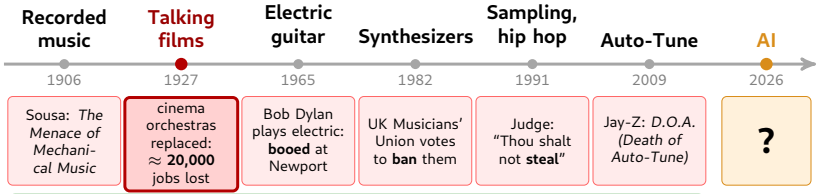
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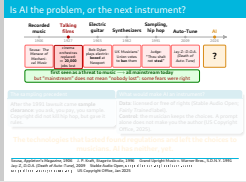
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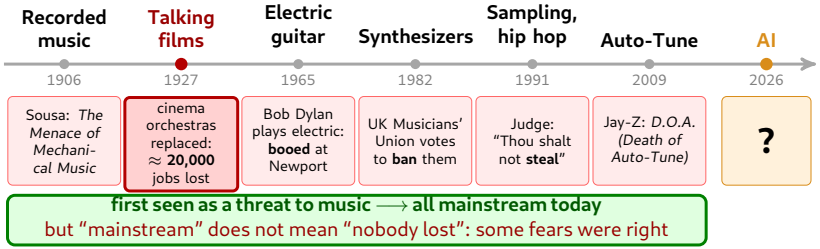
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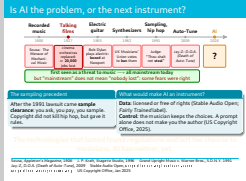
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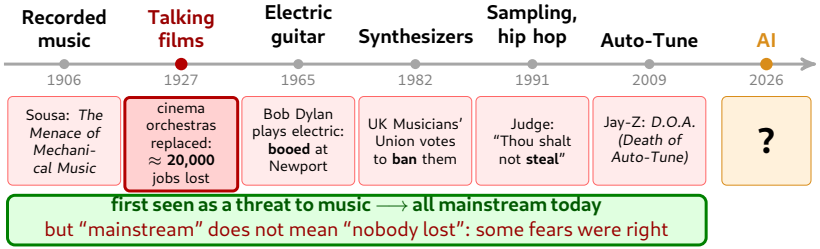
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The sampling precedent

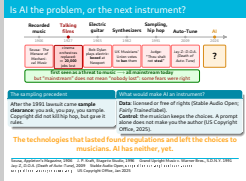
After the 1991 lawsuit came **sample clearance**: you ask, you pay, you sample. Copyright did not kill hip hop, but gave it rules.

What would make AI an instrument?

Data: licensed or free of rights (Stable Audio Open; *Fairly Trained* label).
Control: the musician keeps the choices. A prompt alone does not make you the author (US Copyright Office, 2025).

The technologies that lasted found regulations and left the choices to musicians. AI has neither, yet.

Sousa, *Appleton's Magazine*, 1906 J. P. Kraft, *Stage to Studio*, 1996 Grand Upright Music v. Warner Bros., S.D.N.Y. 1991
Jay-Z, *D.O.A. (Death of Auto-Tune)*, 2009 Stable Audio Open, https://arxiv.org/abs/2407.14358
https://www.fairlytrained.org US Copyright Office, Jan 2025



Sousa made the following argument in 1906: "Why learn to play if a machine plays for you?" This is the same as "why learn music if Suno makes the song?".
The 1991 ruling was against Biz Markie, for a sample used without permission.
Hip hop, built on sampling, has been the most consumed genre in the US since 2017.
In 2009, at the peak of Auto-Tune in rap and R&B, Jay-Z released *D.O.A. (Death of Auto-Tune)*, asking rappers to stop using it. The song won a Grammy, yet Auto-Tune never left: it is now a standard tool in pop and rap production.
This timeline is built from the technologies that *succeeded*: a survivorship bias. "It became normal" and "nobody lost" are two different claims.

Stable Audio Open was trained only on Creative Commons sounds; *Fairly Trained* certifies models trained with consent. According to the US Copyright Office, a purely generated song may have *no* author at all.

Place yourself: Generative AI: Instrument, or parasite for the artists?

pick one side

INSTRUMENT *or* **PARASITE**

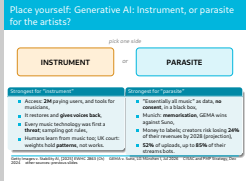
Strongest for "instrument"

- Access: **2M** paying users, and tools for musicians,
- It restores and **gives voices back**,
- Every music technology was first a **threat**; sampling got rules,
- Humans learn from music too; UK court: weights hold **patterns**, not works.

Strongest for "parasite"

- "Essentially all music" as data, **no consent**, in a black box,
- Munich: **memorisation**, GEMA wins against Suno,
- Money to labels; creators risk losing **24%** of their revenues by 2028 (projection),
- **52%** of uploads, up to **85%** of their streams bots.

Getty Images v. Stability AI, [2025] EWHC 2863 (Ch) GEMA v. Suno, LG München I, Jul 2026 CISAC and PMP Strategy, Dec 2024 other sources: previous slides



Choose a side for AI music **as it is used today**, and say which uses you have in mind: the uncontested uses (Lennon's voice, Randy Travis) and the contested ones (cloned voices, fake bands) rely on the same family of technology. The difference is rarely in the tool: it is in the consent, the choices left to the human, and where the money goes.

The UK decision (Getty Images v. Stability AI, 2025) is about an image model: its weights contain patterns learned from the data, not copies of the works.

Two questions to test your position. **Does a licensed model settle it?** Paying rights owners answers a question about *property*, not necessarily about *artists*.

Who was owed something? The label, the performer, the composer, or the musical scene a style came from?

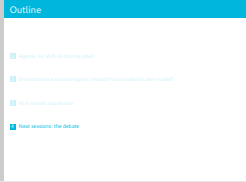
Outline

1 Agentic AI: Will AI destroy jobs?

2 Environment and sovereignty: should France build its own model?

3 AI in artistic expression

4 Next sessions: the debate



Two sessions: prepare, then debate

1. **Work.** Will AI destroy jobs?
   yes ↔ no
2. **Environment and sovereignty.** Should France build its own frontier model?
   yes ↔ no
3. **Artistic expression.** Generative AI: Instrument, or parasite for the artists?
   instrument ↔ parasite

Session 1: become an expert

- You prepare **one** question, **on your own**,
- Read the slides, their notes, **and their sources**; look for new ones,
- Prepare arguments for **both** sides,
- On your question, you have **priority** at the start of the debate to introduce the first arguments.

Session 2: the moving debate

- **20 minutes** per question,
- The room is split in two: one side per answer,
- Experts open, then everyone joins,
- You **move** when an argument convinces you.



We ask you to prepare one question in more detail. Slides are a starting point for your research, but you should go further and look for additional information.

A good argument has a source (or an honest "this is my intuition") and a mechanism. "AI will destroy jobs" is an opinion; "agents now complete tasks of a working day with 50% success, and young people in the most exposed occupations are already hired less" is an argument.

What to avoid: one striking case presented as a general rule, a curve extended to a conclusion, "it is inevitable", "it is just a tool".

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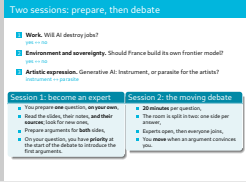
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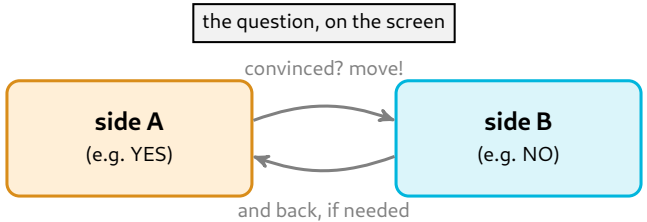


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The rules of the moving debate



Speaking

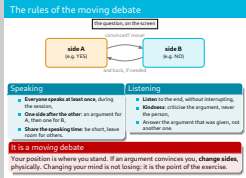
- **Everyone speaks at least once**, during the session,
- **One side after the other**: an argument for A, then one for B,
- **Share the speaking time**: be short, leave room for others.

Listening

- **Listen** to the end, without interrupting,
- **Kindness**: criticise the argument, never the person,
- Answer the argument that was given, not another one.

It is a *moving* debate

Your position is where you stand. If an argument convinces you, **change sides**, physically. Changing your mind is not losing: it is the point of the exercise.



Keep your interventions short (one argument, its source, and that is it): it is what makes it possible for everyone to speak. Kindness is not politeness for its own sake: a debate where people are afraid to speak only hears the most confident voices, not necessarily the best informed.

At the end of each question, look at who moved and why: the arguments that made people move can be very strong ones.